



*International
Virtual
Observatory
Alliance*

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Working Group

Data Models

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Previous versions

This is the first public release

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Abstract

This document defines a data model for stellar occultation observations by small Solar System bodies (asteroids, Trans-Neptunian Objects, Centaurs, Trojans, and comets). Stellar occultations occur when a Solar System body passes in front of a background star as seen from Earth, causing a characteristic drop in the star’s apparent brightness. The resulting light curves encode high-precision information about the occulting body’s size, shape, and any surrounding structures.

The model is designed as an extension to EPN-TAP 2.0 (Europlanet Table Access Protocol), following established IVOA patterns and leveraging existing standards including VO-DML, Coords, and Meas. The source characterization component (OccultedStar) is designed to be compatible with MANGO for epoch propagation support.

The architecture follows a strict 2-class design:

- **OccultationEvent** — Primary class containing event records with embedded star and body parameters
- **OccultationFeature** — Secondary class for auxiliary detections (rings, atmosphere, satellites)

Individual chord observations are accessed via the DataLink protocol, consistent with the ObsCore pattern for accessing related data products.

Status of this document

This is an IVOA Working Draft for review by IVOA members and other interested parties. It is a draft document and may be updated, replaced, or obsoleted by other documents at any time. It is inappropriate to use IVOA Working Drafts as reference materials or to cite them as other than “work in progress”.

A list of current IVOA Recommendations and other technical documents can be found in the IVOA document repository¹.

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¹<https://www.ivoa.net/documents/>

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Conformance-related definitions

The words “MUST”, “SHALL”, “SHOULD”, “MAY”, “RECOMMENDED”, and “OPTIONAL” (in upper or lower case) used in this document are to be interpreted as described in IETF standard RFC2119 (?).

The *Virtual Observatory (VO)* is a general term for a collection of federated resources that can be used to conduct astronomical research, education, and outreach. The **International Virtual Observatory Alliance (IVOA)** is a global collaboration of separately funded projects to develop standards and infrastructure that enable VO applications.

Conformance levels for implementations of this data model:

- **Level 1 (Minimal)**: Implementation of OccultationEvent with mandatory parameters only. Service exposes data via TAP.

- **Level 2 (Standard)**: Level 1 plus OccultationFeature and DataLink endpoint for chord-level data access.
- **Level 3 (Full)**: Level 2 plus MIVOT annotations for MANGO compatibility and visualization product URLs.

1 Introduction

1.1 Role within the VO Architecture

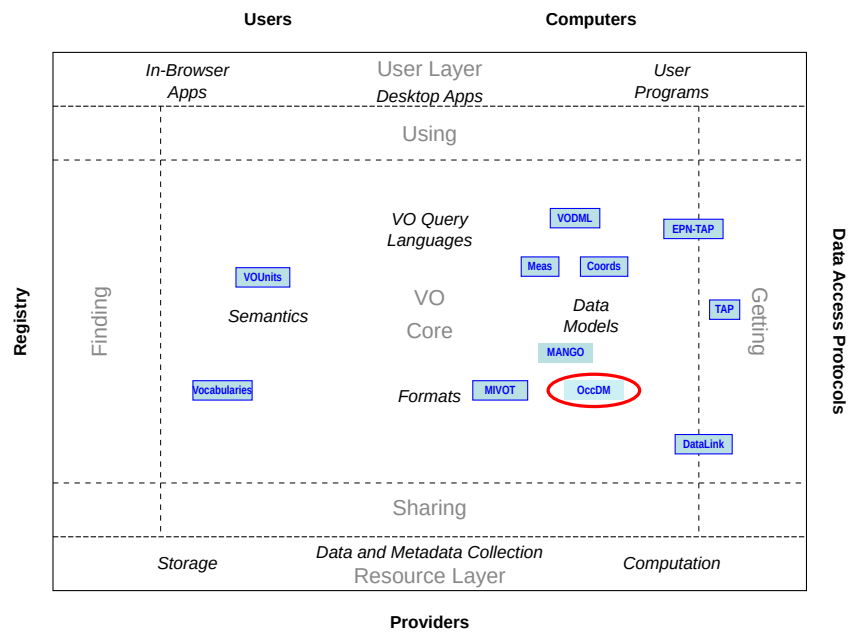


Figure 1: Architecture diagram for this document. OccDM (highlighted) is positioned in the Data Models area, building upon VODML, Meas, and Coords, with MANGO compatibility. It interfaces with TAP for queries and DataLink for chord data access.

Fig. 1 shows the role this document plays within the IVOA architecture (?).

The model relates to other IVOA standards as follows:

- **Base layer (VO-DML)**: Provides the formal modeling framework and primitive types (?).
- **Measurement layer (Coords, Meas)**: Provides coordinate systems and measurement types with errors (??).

- **Source layer (MANGO):** Provides patterns for source characterization (OccultedStar mapping).
- **Discovery layer (EPN-TAP):** Provides the query interface and base parameter vocabulary (?).
- **Access layer (DataLink):** Provides access to chord-level data products (?).
- **Annotation layer (MIVOT):** Enables annotation of VOTable responses for client interpretation (?).

1.2 Motivation

Stellar occultations occur when a Solar System body passes in front of a background star as seen from Earth. The star’s light is blocked, producing a characteristic light curve that encodes information about the occulting body’s size, shape, and any surrounding structures (rings, atmosphere, satellites). These observations provide unique constraints on physical parameters that are difficult to obtain by other means:

- **Size and shape:** Kilometric precision, far exceeding direct imaging capabilities for distant objects
- **Ring systems:** Detection of structures a few kilometers wide at distances of tens of AU
- **Atmospheres:** Sensitivity to nanobar pressure levels, orders of magnitude better than spectroscopy
- **Satellites:** Detection of companions too faint or too close for direct imaging
- **Geometric albedo:** Combined with absolute magnitude, provides albedo without assumptions about shape

Over the past decades, occultation programs have accumulated thousands of observations. These data have led to fundamental discoveries including: the rings of Chariklo (?), the rings of Haumea (?), and the bi-lobed shape of Arrokoth (?).

The problem: Currently, occultation data is scattered across individual publications, personal databases, and institutional archives with no common format. Researchers must contact individual groups and manually convert data formats to combine observations.

The solution: A standardized data model that enables interoperability across archives, federated queries through the Virtual Observatory infrastructure, and systematic analysis across multiple events and targets.

1.3 Context and Scope

This document defines a data model for stellar occultation observations by small Solar System bodies. The scope includes:

- Event-level metadata (timing, geometry, observer configuration)
- Derived physical parameters (size, shape, albedo)
- Secondary feature detections (rings, atmosphere, satellites)
- Individual chord observations (via DataLink)
- Occulted star characterization (compatible with MANGO)
- Occulting body identification and orbital elements

Out of scope: Raw light curve data (FITS files), prediction ephemerides, real-time alert systems, lunar occultations, exoplanet transits.

The model is designed as an extension to EPN-TAP 2.0, the Europlanet Table Access Protocol for planetary science data. It follows the patterns established by ObsCore (?) for observation discovery and DataLink for accessing related data products.

2 Use Cases and Requirements

This section presents the primary use cases that have motivated the development of this data model, followed by the derived requirements.

2.1 Use Cases

2.1.1 UC1: Discovery of All Occultations for a Given Target

Actor: Planetary scientist studying small Solar System objects

Goal: Find all recorded occultation events for a targeted SSO

Scenario: The scientist queries a TAP service with the target name or MPC designation. The service returns all events for that target, including derived parameters (diameter, albedo) and indicators for secondary features (rings, atmosphere).

Derived requirements: Unambiguous target identification (MPC designation), queryable via ADQL, support for name resolution.

2.1.2 UC2: Access to Individual Chord Data

Actor: Researcher performing shape modeling or re-analysis

Goal: Retrieve all chord observations (ingress/egress times, observer locations) for a specific event to perform an independent fit

Scenario: After discovering an event of interest, the researcher follows the DataLink URL to retrieve a VOTable containing all chord records. Each chord includes timing measurements with uncertainties, observer geodetic coordinates, and instrumentation metadata.

Derived requirements: DataLink endpoint, chord timing with errors, observer location, timing source (GPS/NTP).

2.1.3 UC3: Survey of Ring-Bearing Small Bodies

Actor: Researcher studying ring system formation and stability

Goal: Find all small bodies with confirmed or suspected ring detections

Scenario: Query for events where `has_rings = 'yes'` or `'suspected'`. For positive detections, retrieve ring parameters (radius, width, optical depth) from the OccultationFeature class.

Derived requirements: Feature indicators in main class, separate feature class with detailed parameters.

2.1.4 UC4: Temporal Evolution of Atmosphere/Rings

Actor: Researcher monitoring atmospheric evolution (e.g., Pluto, Triton)

Goal: Track changes in atmospheric pressure or ring optical depth across multiple events over years/decades

Scenario: Query all events for a target, filter by feature type, compare derived parameters across epochs.

Derived requirements: Precise event timing (JD UTC), consistent parameter definitions with uncertainties, bibliographic references for traceability.

2.1.5 UC5: Star Position Epoch Propagation

Actor: Prediction service or occultation planner

Goal: Propagate the occulted star's position to the event epoch for shadow path calculation

Scenario: Retrieve star parameters (position, proper motion, parallax, errors) and propagate to `event_epoch` using standard astrometric transformation. MANGO-compatible clients can perform this automatically.

Derived requirements: Complete astrometric parameters with correlated errors, MANGO compatibility for OccultedStar.

2.1.6 UC6: Cross-Match with External Catalogs

Actor: Data center or researcher combining occultation data with other sources

Goal: Link occultation-derived parameters with thermal observations, radar measurements, or spectroscopic data

Scenario: Use MPC designation as primary key to join with external databases (SBDB, Horizons, SsODNet).

Derived requirements: Standardized target identification, alternative designations array, target class vocabulary consistent with EPN-TAP.

2.1.7 UC7: Visualization of Event Results

Actor: Researcher or outreach coordinator

Goal: Access graphical representations of event results

Scenario: Retrieve URLs to pre-generated visualization products showing body limb with chord projections and terrestrial map of observer station distribution.

Derived requirements: URL fields for visualization products.

2.2 Requirements

Requirements are classified as MUST (essential), SHOULD (recommended), or MAY (optional). The reference to motivating use cases is indicated in brackets.

2.2.1 General Requirements

[R1] [MUST] The model SHALL be expressible in VO-DML and serializable to VO-DML/XML.

[R2] [MUST] The model SHALL reuse existing IVOA primitive types (ivoa:string, ivoa:real, ivoa:RealQuantity).

[R3] [MUST] The model SHALL import coords and meas models for coordinate and measurement types.

[R4] [MUST] The model SHALL be implementable as an EPN-TAP extension. [Stakeholder constraint]

[R5] [SHOULD] The model SHOULD provide MANGO-compatible source characterization for OccultedStar. [UC5]

2.2.2 Application Requirements

[R6] [MUST] Services implementing this model SHALL expose data via TAP with ADQL query support. [UC1, UC3]

- [R7] [MUST] Services SHALL provide DataLink endpoints for chord-level data access. [UC2]
- [R8] [SHOULD] Services SHOULD provide MIVOT annotations for MANGO-compatible clients. [UC5]
- [R9] [MUST] Services SHALL register in the IVOA Registry with appropriate resource types. [UC6]
- [R10] [MAY] Services MAY provide visualization product URLs. [UC7]

2.2.3 Content Requirements

- [R11] [MUST] Event records SHALL include MPC packed designation for unambiguous body identification. [UC1, UC6]
- [R12] [MUST] Event records SHALL include sufficient stellar parameters to enable epoch propagation. [UC5]
- [R13] [MUST] Timing measurements SHALL be expressed in JD (UTC) with stated precision. [UC2, UC4]
- [R14] [MUST] Derived parameters SHALL include associated uncertainties. [UC4]
- [R15] [MUST] Chord data SHALL include observer geodetic coordinates and timing source. [UC2]
- [R16] [MUST] All numerical parameters SHALL specify unit and number of decimal places. [All]

3 Model: StellarOccultation

3.1 Model Identification

Property	Value
name	StellarOccultation
version	1.0
uri	ivo://ivoa.net/dm/StellarOccultation
author	T. Chope (LIRA/Observatoire de Paris)
lastModified	2026-01-16

Table 1: Model identification

3.2 Model Imports

This model imports the following IVOA models:

Model	Version	Usage
ivoa	1.0	Primitive types (string, real, boolean, RealQuantity, anyURI)
coords	1.0	LonLatPoint (observer geodetic position)
meas	2.0	Time, Position, ProperMotion, GenericMeasure (with errors)

Table 2: Model imports

3.3 Type Summary

ObjectTypes (classes that may be persisted):

- **OccultationEvent** — Primary class containing occultation event records
- **OccultationFeature** — Secondary class for auxiliary detections

DataTypes (embedded or accessed via DataLink):

- **OccultedStar** — Background star parameters (embedded in OccultationEvent)
- **OccultingBody** — Small body parameters (embedded in OccultationEvent)
- **ChordData** — Individual chord observation (accessed via DataLink, NOT persisted)

Enumerations:

- **TargetClass** — asteroid | TNO | Centaur | dwarf_planet | comet | satellite | Trojan
- **StatusFlag** — yes | no | suspected
- **DuplicityFlag** — single | binary | suspected_binary | multiple | unknown
- **ChordStatus** — positive | negative | grazing
- **TimingSource** — GPS | NTP | manual | unknown
- **FeatureType** — ring | atmosphere | satellite
- **DetectionType** — detection | upper_limit | marginal

Figure 2: VO-DML class diagram of the Stellar Occultation Data Model. Two ObjectTypes (green), three DataTypes (blue for embedded, cyan for DataLink-accessed), seven enumerations (yellow), model imports (purple), and MANGO compatibility annotations.

3.4 Architecture Diagram

4 OccultationEvent

The primary ObjectType representing a stellar occultation event. One instance represents exactly one event: a single small body occulting a single star at a specific epoch.

4.1 Event Identification

4.1.1 event_id

vodml-id
OccultationEvent.event_id
type
ivoa:string
multiplicity
1 (mandatory)

Unique identifier for this occultation event within the service. The recommended format is YYYYMMDD_TARGETNAME where TARGETNAME is the MPC packed designation (e.g., 20240610_K03M12W for Varda). This identifier serves as the primary key and is used for DataLink chord access queries.

4.1.2 event_epoch

vodml-id
OccultationEvent.event_epoch
type
meas:Time
multiplicity
1 (mandatory)

The geocentric closest approach time in Julian Date (UTC). This is the nominal occultation time, typically corresponding to the midpoint of the central chord or the predicted closest approach if no central chord was observed. The meas:Time type bundles a coords:TimeStamp with measurement errors. Precision: 0.1 second (JD with 6 decimal places).

4.1.3 shadow_velocity

vodml-id
OccultationEvent.shadow_velocity
type
meas:GenericMeasure
multiplicity
1 (mandatory)

The velocity of the shadow across the Earth's surface in km/s. This is the apparent velocity combining the body's orbital motion, Earth's rotation, and parallax effects. Negative values indicate retrograde motion. This parameter is critical for converting timing measurements to physical distances ($\text{chord_length} = \text{shadow_velocity} \times \Delta t$). Unit: km/s, precision: 2 decimal places.

4.1.4 shadow_pa

vodml-id
OccultationEvent.shadow_pa
type
ivoa:RealQuantity
multiplicity
0..1

Position angle of the shadow motion direction, measured East of North. Unit: degrees, precision: 1 decimal place.

4.1.5 closest_approach

vodml-id
OccultationEvent.closest_approach
type
ivoa:RealQuantity
multiplicity
1 (mandatory)

The geocentric closest approach distance between the body center and the star, expressed as an angular separation. Unit: arcsec, precision: 3 decimal places.

4.1.6 event_duration

vodml-id
OccultationEvent.event_duration
type
meas:GenericMeasure
multiplicity
0..1

Maximum duration of the occultation (for the central chord if available).
Unit: seconds, precision: 1 decimal place.

4.2 Chord Statistics

4.2.1 nb_chords_positive

vodml-id
OccultationEvent.nb_chords_positive
type
ivoa:nonnegativeInteger
multiplicity
1 (mandatory)

Number of positive chord detections (observers who recorded a clear occultation with both ingress and egress).

4.2.2 nb_chords_negative

vodml-id
OccultationEvent.nb_chords_negative
type
ivoa:nonnegativeInteger
multiplicity
0..1

Number of negative observations (observers who did not detect the occultation despite adequate coverage).

4.2.3 nb_chords_grazing

vodml-id
OccultationEvent.nb_chords_grazing
type
ivoa:nonnegativeInteger

multiplicity

0..1

Number of grazing chords (partial occultations near the limb edge).

4.2.4 datalink_url

vodml-id

OccultationEvent.datalink_url

type

ivoa:anyURI

multiplicity

1 (mandatory)

URL to the DataLink service endpoint for accessing individual chord observations. Query pattern: `GET {datalink_url}?ID={event_id}` returns a VOTable with ChordData structure. The service follows IVOA DataLink 1.1 protocol.

4.3 Derived Physical Parameters

Parameters derived from the occultation fit. These represent the primary scientific output and require at least one positive chord.

4.3.1 equiv_diameter

vodml-id

OccultationEvent.equiv_diameter

type

meas:GenericMeasure

multiplicity

0..1

The area-equivalent circular diameter: $D = 2\sqrt{A/\pi}$ where A is the projected area from the ellipse fit. Unit: km, precision: 1 decimal place.

4.3.2 ellipse_a

vodml-id

OccultationEvent.ellipse_a

type

meas:GenericMeasure

multiplicity

0..1

Semi-major axis of the best-fit ellipse to the limb points. Unit: km, precision: 1 decimal place.

4.3.3 ellipse_b

vodml-id
OccultationEvent.ellipse_b
type
meas:GenericMeasure
multiplicity
0..1

Semi-minor axis of the best-fit ellipse. Unit: km, precision: 1 decimal place.

4.3.4 ellipse_pa

vodml-id
OccultationEvent.ellipse_pa
type
ivoa:RealQuantity
multiplicity
0..1

Position angle of the ellipse major axis, measured East of North in the sky plane. Unit: degrees, precision: 1 decimal place.

4.3.5 geometric_albedo

vodml-id
OccultationEvent.geometric_albedo
type
meas:GenericMeasure
multiplicity
0..1

Geometric albedo derived from the occultation diameter combined with the absolute magnitude. Calculated using: $p_v = (1329/D)^2 \times 10^{-0.4 \times H}$ where D is equiv_diameter in km and H is the absolute magnitude. Dimensionless, precision: 2 decimal places.

4.3.6 oblateness

vodml-id
OccultationEvent.oblateness
type
ivoa:real
multiplicity
0..1

Oblateness defined as $(a - b)/a$ where a and b are the semi-major and semi-minor axes. Dimensionless, precision: 2 decimal places.

4.4 Secondary Feature Indicators

Boolean-like flags indicating the presence of secondary structures. When set to 'yes', corresponding detailed parameters are available in the Occultation-Feature class.

4.4.1 has_rings

vodml-id

OccultationEvent.has_rings

type

StatusFlag

multiplicity

0..1

Indicates ring detection. Values: 'yes' (confirmed), 'no' (ruled out with upper limits), 'suspected' (marginal detection).

4.4.2 has_atmosphere

vodml-id

OccultationEvent.has_atmosphere

type

StatusFlag

multiplicity

0..1

Indicates atmosphere detection. Values: 'yes' (confirmed), 'no' (ruled out), 'suspected' (marginal detection).

4.4.3 has_satellite

vodml-id

OccultationEvent.has_satellite

type

StatusFlag

multiplicity

0..1

Indicates satellite detection via secondary occultation event.

4.5 Metadata

4.5.1 bib_reference

vodml-id
OccultationEvent.bib_reference
type
ivoa:string
multiplicity
0..1

Bibcode or DOI of the primary publication describing this event (e.g., 2020A&A...643A.125S for Varda).

4.5.2 timing_source

vodml-id
OccultationEvent.timing_source
type
TimingSource
multiplicity
1 (mandatory)

Timestamp synchronization method: 'GPS' (best), 'NTP', 'manual', 'unknown'. Critical for assessing timing accuracy.

4.6 Compositions

OccultationEvent contains two embedded DataTypes:

4.6.1 star : OccultedStar [1]

Composition of the occulted star characterization. See Section 5.

4.6.2 target : OccultingBody [1]

Composition of the occulting body characterization. See Section 6.

5 OccultedStar

DataType for background star characterization. Embedded in Occultation-Event via composition. MANGO-compatible for epoch propagation support.

5.1 Star Identification

5.1.1 star_id

vodml-id
OccultedStar.star_id
type
ivoa:string
multiplicity
1 (mandatory)

Unique identifier in the reference catalog, typically Gaia DR3 source_id (e.g., 4367203805493754752).

5.1.2 star_catalogue

vodml-id
OccultedStar.star_catalogue
type
ivoa:string
multiplicity
1 (mandatory)

Reference catalog name. Recommended values: 'GaiaDR3', 'GaiaDR2', 'UCAC4', 'Tycho-2'.

5.2 Astrometry

5.2.1 position

vodml-id
OccultedStar.position
type
meas:Position
multiplicity
1 (mandatory)

Star position at catalog reference epoch (typically J2016.0 for Gaia DR3). The meas:Position type bundles coords:LonLatPoint with errors and correlations. Precision: 6 decimal places for degrees (~ 3.6 mas).

5.2.2 proper_motion

vodml-id
OccultedStar.proper_motion
type
meas:ProperMotion
multiplicity
0..1

Proper motion in RA and Dec. Unit: mas/yr, precision: 3 decimal places.

5.2.3 star_ra_error, star_dec_error

vodml-id
OccultedStar.star_ra_error / star_dec_error
type
ivoa:RealQuantity
multiplicity
0..1

Position uncertainties. Unit: mas, precision: 2 decimal places.

5.2.4 parallax

vodml-id
OccultedStar.parallax
type
ivoa:RealQuantity
multiplicity
0..1

Trigonometric parallax. Unit: mas, precision: 3 decimal places.

5.3 Photometry

5.3.1 star_mag_G, star_mag_RP, star_mag_BP

vodml-id
OccultedStar.star_mag_G / RP / BP
type
ivoa:real
multiplicity
G: 1, RP/BP: 0..1

Gaia photometry bands. star_mag_G is mandatory as it determines the SNR requirements. Precision: 2 decimal places.

5.3.2 `star_mag_J`, `star_mag_H`, `star_mag_K`

`vodml-id`

`OccultedStar.star_mag_J / H / K`

`type`

`ivoa:real`

`multiplicity`

`0..1`

2MASS infrared magnitudes. Precision: 2 decimal places.

5.4 Quality Indicators

5.4.1 `star_ruwe`

`vodml-id`

`OccultedStar.star_ruwe`

`type`

`ivoa:real`

`multiplicity`

`0..1`

Gaia Renormalized Unit Weight Error. $\text{RUWE} > 1.4$ may indicate binarity or astrometric problems. Precision: 2 decimal places.

5.4.2 `star_duplicity`

`vodml-id`

`OccultedStar.star_duplicity`

`type`

`DuplicityFlag`

`multiplicity`

`0..1`

Stellar multiplicity status. Values: 'single', 'binary', 'suspected_binary', 'multiple', 'unknown'.

5.4.3 `angular_diameter`

`vodml-id`

`OccultedStar.angular_diameter`

`type`

`meas:GenericMeasure`

`multiplicity`

`0..1`

Estimated stellar angular diameter (from color relations). Unit: mas, precision: 2 decimal places.

5.4.4 spectral_type

vodml-id
OccultedStar.spectral_type
type
ivoa:string
multiplicity
0..1

MK spectral classification if known (e.g., 'K2III').

6 OccultingBody

DataType for occulting body characterization. Embedded in Occultation-Event via composition.

6.1 Target Identification

6.1.1 target_name

vodml-id
OccultingBody.target_name
type
ivoa:string
multiplicity
1 (mandatory)

MPC packed designation (e.g., K03M12W for (174567) Varda). This is the primary key for cross-matching. See <https://minorplanetcenter.net/iau/info/PackedDes.html>.

6.1.2 target_name_readable

vodml-id
OccultingBody.target_name_readable
type
ivoa:string
multiplicity
0..1

Human-readable designation (e.g., (174567) Varda).

6.1.3 target_class

vodml-id
OccultingBody.target_class
type
TargetClass
multiplicity
1 (mandatory)

Body classification. Values: 'asteroid', 'TNO', 'Centaur', 'dwarf_planet', 'comet', 'satellite', 'Trojan'. Based on EPN-TAP vocabulary.

6.1.4 dynamical_class

vodml-id
OccultingBody.dynamical_class
type
ivoa:string
multiplicity
0..1

Detailed dynamical classification (e.g., `classical-hot`, `resonant-3:2`). This value is frozen at event_epoch.

6.2 Physical Properties

6.2.1 target_geocentric_distance

vodml-id
OccultingBody.target_geocentric_distance
type
ivoa:RealQuantity
multiplicity
1 (mandatory)

Distance from Earth at event epoch. Unit: AU, precision: 5 decimal places.

6.2.2 ephemeris_source

vodml-id
OccultingBody.ephemeris_source
type
ivoa:string

multiplicity

0..1

Ephemeris used for predictions (e.g., NIMA^{v13}, JPL#45).

6.3 Orbital Elements

6.3.1 semi_major_axis

vodml-id

OccultingBody.semi_major_axis

type

ivoa:RealQuantity

multiplicity

0..1

Semi-major axis of the heliocentric orbit. Unit: AU, precision: 2 decimal places.

6.3.2 eccentricity

vodml-id

OccultingBody.eccentricity

type

ivoa:real

multiplicity

0..1

Orbital eccentricity. Dimensionless, precision: 3 decimal places.

6.3.3 inclination

vodml-id

OccultingBody.inclination

type

ivoa:RealQuantity

multiplicity

0..1

Orbital inclination to the ecliptic. Unit: degrees, precision: 1 decimal place.

7 ChordData (via DataLink)

DataType for individual chord observations. Accessed via DataLink protocol, NOT persisted as a database table. Each chord represents one observer's recording of the occultation event.

7.1 Chord Identification

7.1.1 chord_id

vodml-id
ChordData.chord_id
type
ivoa:string
multiplicity
1 (mandatory)

Unique identifier for this chord within the event. Recommended format: `event_id_NNN` (sequential numbering).

7.1.2 parent_event_id

vodml-id
ChordData.parent_event_id
type
ivoa:string
multiplicity
1 (mandatory)

Reference to the parent `OccultationEvent.event_id`.

7.2 Timing Measurements

7.2.1 ingress_time

vodml-id
ChordData.ingress_time
type
meas:Time
multiplicity
0..1

Time when the star disappears (start of occultation). Unit: JD (UTC), precision: 0.001 second. May be absent for negative chords.

7.2.2 egress_time

vodml-id
ChordData.egress_time
type
meas:Time

multiplicity

0..1

Time when the star reappears (end of occultation). Unit: JD (UTC), precision: 0.001 second.

7.2.3 chord_length

vodml-id

ChordData.chord_length

type

meas:GenericMeasure

multiplicity

0..1

Length of the chord calculated from timing and shadow velocity. Unit: km, precision: 1 decimal place.

7.2.4 chord_status

vodml-id

ChordData.chord_status

type

ChordStatus

multiplicity

1 (mandatory)

Quality flag: 'positive' (full occultation), 'negative' (no detection), 'grazing' (partial).

7.3 Observer Location

7.3.1 observer_location

vodml-id

ChordData.observer_location

type

coords:LonLatPoint

multiplicity

1 (mandatory)

Geodetic position of the observing station. Longitude East positive, latitude North positive. Precision: 5 decimal places (~ 1 m).

7.3.2 observer_altitude

vodml-id ChordData.observer_altitude
type ivoa:RealQuantity
multiplicity 0..1

Altitude above WGS84 ellipsoid. Unit: meters, precision: 1 decimal place.

7.4 Instrumentation

7.4.1 telescope_aperture

vodml-id ChordData.telescope_aperture
type ivoa:RealQuantity
multiplicity 0..1

Telescope aperture diameter. Unit: meters, precision: 2 decimal places.

7.4.2 timing_source

vodml-id ChordData.timing_source
type TimingSource
multiplicity 1 (mandatory)

Timestamp synchronization method for this specific station.

7.4.3 snr

vodml-id ChordData.snr
type ivoa:real
multiplicity 0..1

Signal-to-noise ratio of the detection. Dimensionless.

8 OccultationFeature

ObjectType for auxiliary detections (rings, atmosphere, satellites). Related to OccultationEvent via aggregation (one event may have multiple features).

8.1 Feature Identification

8.1.1 feature_id

vodml-id

OccultationFeature.feature_id

type

ivoa:string

multiplicity

1 (mandatory)

Unique identifier for this feature. Recommended format: `event_id_type_NNN`.

8.1.2 parent_event_id

vodml-id

OccultationFeature.parent_event_id

type

ivoa:string

multiplicity

1 (mandatory)

Reference to OccultationEvent.event_id. This establishes the aggregation relationship.

8.1.3 feature_type

vodml-id

OccultationFeature.feature_type

type

FeatureType

multiplicity

1 (mandatory)

Discriminator: 'ring', 'atmosphere', 'satellite'. Determines which conditional parameters apply.

8.1.4 detection_type

vodml-id
OccultationFeature.detection_type
type
DetectionType
multiplicity
1 (mandatory)

Quality flag: 'detection' (confirmed), 'upper_limit' (non-detection with constraint), 'marginal' (tentative).

8.2 Ring Parameters (feature_type = 'ring')

8.2.1 ring_name

vodml-id
OccultationFeature.ring_name
type
ivoa:string
multiplicity
0..1

Ring designation (e.g., 'C1R', 'Q2R' for Chariklo/Quaoar rings).

8.2.2 ring_radius

vodml-id
OccultationFeature.ring_radius
type
meas:GenericMeasure
multiplicity
0..1

Mean orbital radius from body center. Unit: km, precision: 1 decimal place.

8.2.3 ring_width

vodml-id
OccultationFeature.ring_width
type
meas:GenericMeasure
multiplicity
0..1

Radial width of the ring. Unit: km, precision: 1 decimal place.

8.2.4 ring_optical_depth

vodml-id
OccultationFeature.ring_optical_depth
type
meas:GenericMeasure
multiplicity
0..1

Normal optical depth (τ). Dimensionless, precision: 2 decimal places.

8.3 Atmosphere Parameters (feature_type = 'atmosphere')

8.3.1 half_light_radius

vodml-id
OccultationFeature.half_light_radius
type
meas:GenericMeasure
multiplicity
0..1

Radius at 50% flux level. Unit: km, precision: 1 decimal place.

8.3.2 surface_pressure

vodml-id
OccultationFeature.surface_pressure
type
meas:GenericMeasure
multiplicity
0..1

Estimated surface pressure. Unit: μ bar, precision: 2 decimal places.

8.3.3 scale_height

vodml-id
OccultationFeature.scale_height
type
meas:GenericMeasure
multiplicity
0..1

Atmospheric scale height. Unit: km, precision: 1 decimal place.

9 Open Issues and Future Work

9.1 Dynamical Class Evolution

Some objects have dynamical classifications that evolve due to gravitational perturbations. Example: (99942) Apophis transitions from “Aten” to “Apollo” class after its 2029 Earth encounter.

Current solution: The `dynamical_class` field is frozen at `event_epoch`. Future versions may include a time-dependent classification service reference.

9.2 Binary Star Treatment

Binary stars produce complex light curves with potential for multiple events. The current model provides `star_duplicity` as a warning flag. Future versions may extend `OccultedStar` to handle resolved binary parameters.

9.3 Contact Binary Bodies

How to report on contact binaries (e.g., Arrokoth)? The current ellipse parameterization may not adequately describe bi-lobed shapes. Future work: Investigation of shape model parameters.

9.4 MPC Designation Resolution

Integration with MPC name resolver services for automatic designation cross-matching. Reference: https://minorplanetcenter.net/web_service.

A Enumeration Values

A.1 TargetClass

`asteroid` | `TNO` | `Centaur` | `dwarf_planet` | `comet` | `satellite` | `Trojan`

Based on EPN-TAP `target_class` vocabulary.

A.2 StatusFlag

`yes` | `no` | `suspected`

For `has_rings`, `has_atmosphere`, `has_satellite` indicators.

A.3 DuplicityFlag

`single` | `binary` | `suspected_binary` | `multiple` | `unknown`

Stellar multiplicity classification.

A.4 ChordStatus

positive | negative | grazing

positive = full occultation; negative = no detection; grazing = partial.

A.5 TimingSource

GPS | NTP | manual | unknown

Timestamp synchronization method. GPS preferred.

A.6 FeatureType

ring | atmosphere | satellite

Discriminator for OccultationFeature.

A.7 DetectionType

detection | upper_limit | marginal

Quality of feature detection.

B ADQL Query Examples

B.1 Find all occultations for a target

```
SELECT event_id, event_epoch, equiv_diameter, geometric_albedo
FROM occ_event
WHERE target_name = 'K03M12W'
ORDER BY event_epoch
```

B.2 Find ring-bearing objects

```
SELECT DISTINCT e.target_name_readable, f.ring_radius, f.ring_width
FROM occ_event e
JOIN occ_feature f ON e.event_id = f.parent_event_id
WHERE e.has_rings = 'yes' AND f.feature_type = 'ring'
```

B.3 Atmospheric pressure evolution

```
SELECT e.event_epoch, f.surface_pressure
FROM occ_event e
JOIN occ_feature f ON e.event_id = f.parent_event_id
WHERE e.target_name = 'J34P00I' -- Pluto
      AND f.feature_type = 'atmosphere'
ORDER BY e.event_epoch
```

C Changes from Previous Versions

No previous versions yet.